

Educating Leaders for AI-Enabled Decision Environments: Data Architecture, Human Judgment, and Responsible AI Integration

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Proposal Summary

Artificial Intelligence systems are rapidly entering education, business, healthcare, defense, and public-sector decision environments. Many organizational leaders are now being asked to adopt, fund, or operationalize AI capabilities without fully understanding the architectural, human, and governance requirements necessary to produce trustworthy outcomes.

Current discussions surrounding AI often focus on model performance and automation potential. Far less attention is given to the quality of the underlying data systems, the non-deterministic nature of Agentic AI and Large Language Models (LLMs), and the human factors required to maintain responsible decision-making.

This proposal seeks to explore how educational institutions can better prepare current and future leaders to critically evaluate AI-enabled systems and understand the relationship between data architecture, human judgment, and operational outcomes.

The work will place particular emphasis on well-curated data systems where data is governed, trustworthy, credible, traceable, and explainable. The research will investigate how these foundations influence trust, situational awareness, decision quality, and organizational effectiveness when humans interact with AI systems.

The goal is not to position AI as a replacement for human expertise. The goal is to educate leaders to better understand what they are asking AI systems to do, the limitations of those systems, and the conditions required for responsible implementation.

Core Research Questions

1. How should leaders evaluate whether an AI-enabled system is operating on trustworthy and explainable data?
2. What risks emerge when organizations deploy AI systems on fragmented, poorly governed, or incomplete data architectures?
3. How can educational institutions prepare leaders to make informed decisions regarding AI adoption and operational use?
4. What human factors most influence trust, situational awareness, and decision effectiveness in AI-enabled environments?
5. How do organizations maintain human accountability and decision authority when AI systems are introduced into operational workflows?

6. What architectural and governance patterns produce reliable, traceable, and explainable operational outcomes?

Research Motivation

There is growing pressure across industries to rapidly adopt AI technologies. Many leaders are making strategic and operational decisions regarding AI without fully understanding:

- the limitations of non-deterministic AI systems,
- the importance of governed data architectures,
- the role of human oversight,
- or the downstream consequences of opaque automation.

This creates significant educational and operational challenges.

The proposed research argues that responsible AI adoption requires more than technical implementation. It requires leaders who understand:

- how AI systems derive outputs,
- how data quality influences outcomes,
- how explainability and traceability affect trust,
- and how human judgment must remain central in high-consequence environments.

This does not guarantee that every decision will be correct. It does ensure decisions can be justified, explained, and evaluated within a structured operational framework.

Proposed Areas of Investigation

1. AI Literacy for Organizational Leaders

Develop frameworks that help leaders understand AI limitations, risks, governance requirements, and decision implications.

2. Data Architecture as Decision Infrastructure

Examine how governed data systems influence organizational trust, explainability, and AI effectiveness.

3. Human Factors in AI-Enabled Decision Systems

Study trust calibration, cognitive overload, situational awareness, and decision authority in human-AI environments.

4. Responsible AI Education Models

Develop interdisciplinary curriculum concepts bridging engineering, psychology, education, data science, and organizational leadership.

Potential Outcomes

- Educational frameworks for responsible AI leadership
- Executive and graduate-level curriculum development
- Research publications and conference presentations
- Cross-disciplinary collaboration between education, engineering, psychology, and data science
- Practical guidance for organizations implementing AI-enabled systems
- External research and partnership opportunities

Alignment with Jacobs Institute Mission

This proposal aligns directly with the Jacobs Institute's focus on innovation in education and preparing leaders for emerging societal and technological challenges.

The work emphasizes interdisciplinary learning, responsible technology integration, and the development of educational models that help leaders critically evaluate AI systems rather than simply adopt them.

The proposal bridges engineering, human performance, decision science, and education to address one of the most important emerging challenges facing institutions today: educating leaders who understand both the capabilities and limitations of AI-enabled systems.

Researcher Background

Dr. Jonathan Sixt brings over 25 years of experience leading engineering and data platform development in defense and medical technology environments.

His work has focused on enabling high-consequence decision-making through engineered data systems supporting aviation operations, maritime domain awareness, and clinical decision support systems.

His academic background in performance psychology and human decision-making complements ongoing industry experience in data architecture, AI enablement, and operational system integration.